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in a *bomb calorimeter*.

The inner vessel (the bomb) and its cover are made of strong steel. The cover is fitted tightly to the vessel by means of metal lid and screws.

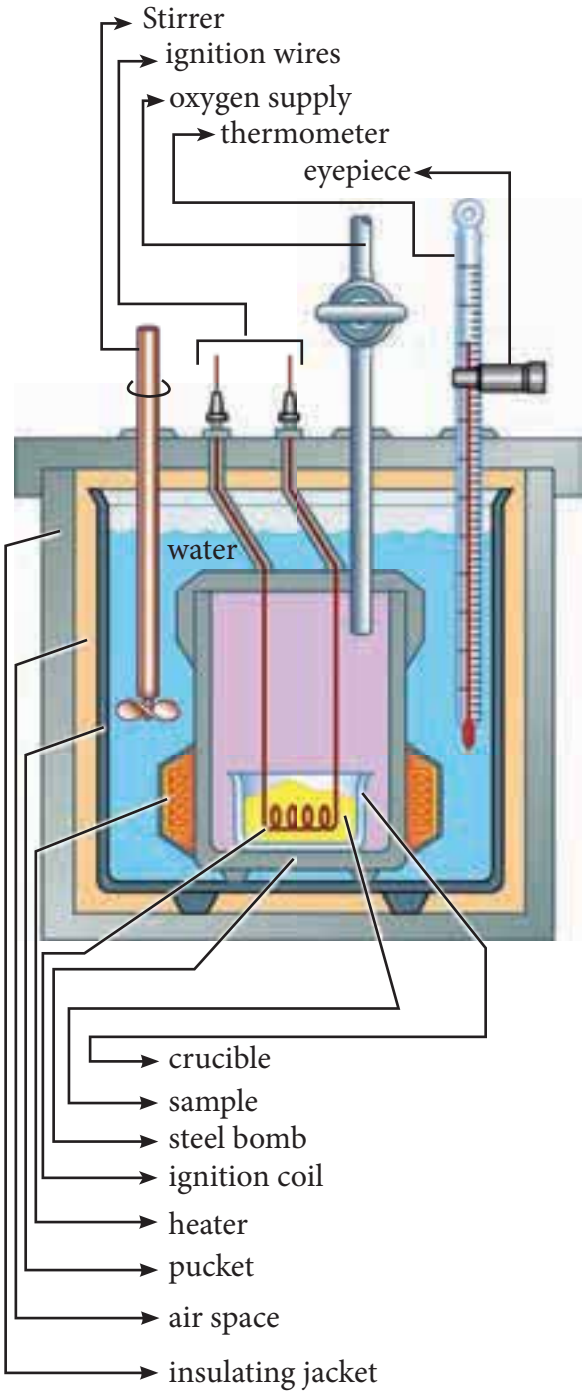


Figure 7.6 Bomb calorimeter

A weighed amount of the substance is taken in a platinum cup connected with electrical wires for striking an arc instantly to kindle combustion. The bomb is then tightly closed and pressurized with excess oxygen. The bomb is immersed in water, in the inner volume of the calorimeter. A stirrer is placed in the space between the wall of the calorimeter and the bomb, so that water can be stirred, uniformly. The reaction is started by striking the substance through electrical heating.

A known amount of combustible substance is burnt in oxygen in the bomb. Heat evolved during the reaction is absorbed by the calorimeter as well as the water in which the bomb is immersed. The change in temperature is measured using a Beckman thermometer. Since the bomb is sealed its volume does not change and hence the heat measurements is equal to the heat of combustion at a constant volume $(\Delta U)_c$.

The amount of heat produced in the reaction $(\Delta U)_c$ is equal to the sum of the heat abosrbed by the calorimeter and water.

Heat absorbed by the calorimeter

$$q_1 = k.\Delta T$$

where k is a calorimeter constant equal to $m_c\ C_c$ (m_c is mass of the calorimeter and C_c is heat capacity of calorimeter)

Heat absorbed by the water

$$q_2 = m_w\ C_w\ \Delta T$$

where m_w is molar mass of water

C_w is molar heat capacity of water (4,184 kJ K^{-1} mol^{-1})